POLI3148: Data Science in Politics and Public Administration

Assignment 1: Data-Driven Analysis of Armed Conflict Using ACLED

Instructor: Prof. Haohan Chen

Weight: 20% of the final course grade

Due Date: April 26, 2026, 11:59 PM

Overview

In this assignment, you will load, clean, visualize, and analyze real-world political data from the **Armed Conflict Location & Event Data Project (ACLED)** with the assistance of an AI coding assistant. The objective is to put the data science skills you have learned in POLI3148 into practice by producing a self-contained data science product that you can showcase as evidence of your analytical capabilities in future career opportunities.

About the ACLED Dataset

ACLED is a disaggregated data collection, analysis, and crisis mapping project that records political violence and protest events worldwide. Maintained by a registered non-profit organization and originally founded by Professor Clionadh Raleigh, ACLED codes the dates, actors, locations, fatalities, and types of events—including battles, violence against civilians, explosions and remote violence, riots, protests, and strategic developments. The dataset covers over 250 countries and territories from 1997 to the present, making it one of the most comprehensive publicly available conflict datasets. It is widely used in political science research because it enables disaggregated, subnational analysis of conflict dynamics, allowing scholars to examine the causes, patterns, and consequences of political violence and social unrest at a granular level.

Deliverables

Your submission should be organized as a **data project folder** containing the following components:

1. Jupyter Notebook(s) for Data Analysis (Required)

Create one or more Jupyter notebooks in which you perform your data analysis step by step. Your notebooks must contain **replicable code** and **clear documentation** explaining what each section accomplishes.

For reference, consult the in-class demonstrations and previous in-class exercises. A well-constructed notebook features code that can be executed from top to bottom without errors, accompanied by informative markdown cells that guide the reader through your analytical process.

2. Interactive Data-Driven Analysis Report (Required)

Based on your analyses in the Jupyter notebooks, produce an **HTML webpage** that presents a self-contained, interactive, data-driven analysis report.

For inspiration, review the reports available under ACLED's Global Analysis section. The following examples illustrate the type of analysis you might aim for:

- **Global scope:** What's Driving Conflict Today: Review of Global Trends
- **Regional scope:** Exhausted Ukraine Faces Military and Diplomatic Pressure to Cede Donbas

These existing reports may serve as inspiration and a point of departure. However, **do not copy any existing report**. Your report must contain original content. If you reference an existing report, be sure to cite it properly. You may begin your project as a replication and extension of an existing report, but to achieve a high grade, your report must demonstrate substantive original analysis.

The target length of the report is approximately **1,500 words** (plus or minus 200 words is acceptable). This word count refers to the analytical text of the report and does not include code, figure captions, or references.

Important note: The interactive data-driven analysis report is not a data visualization dashboard with some supplementary text. The core of this deliverable is an **analytical report** — a coherent, written narrative that poses research questions, describes methodology, presents findings, and draws conclusions. Data visualizations and interactive elements should serve to support and enrich the analysis, not replace it.

3. Note on AI Use (Required)

Submit a note (in Markdown or PDF format, included as a separate file in your project folder) in which you concisely describe how you used an AI coding assistant to complete this assignment. A recommended length is approximately **500 words**, though there is no strict limit. You may include figures and/or tables to illustrate your workflow.

4. README File (Required)

Create a `README.md` file that serves as the **overview document** for your entire project. The README is distinct from the analysis report itself: while the report presents your in-depth analysis, the README provides a high-level summary that orients readers to the project as a whole. When your project is published online (e.g., on GitHub), the README is the first piece of information visitors will see before exploring any of the project files or reading the full report. A well-written README helps others quickly understand what the project is about, how it was done, and what was found.

Your README should cover the following: a description of the data sources used, an overview of the methodology employed, a summary of the key findings, a discussion of the limitations of the findings, and the author's information.

5. Online Publication (Bonus)

Push your data project folder to **GitHub** and make your interactive data report publicly accessible as a webpage via `github.io` (e.g., `https://yourusername.github.io/your-repo/`). To enable this, place your HTML report in the `docs/` folder and configure GitHub Pages to serve from that directory.

Recommended Project Folder Structure

```
project_folder/
+-- README.md
+-- data/
|   +-- acled_data.csv             # Raw data file(s)
+-- code/
|   +-- 01_data_cleaning.ipynb     # Data loading and cleaning
|   +-- 02_exploration.ipynb       # Exploratory data analysis
```

```
|   +-- 03_analysis.ipynb           # Main analysis and visualization
|   +-- Z_generate_report.py        # Python script to generate the report
+-- docs/
|   +-- index.html                  # Interactive data-driven report
+-- note_on_ai_use.md               # Note on AI use
```

You may adapt this structure to suit your project, but your folder should be logically organized with a clear separation between data, code, and outputs. Note that the `docs/` folder is used so that GitHub Pages can directly serve your report as a public webpage.

Getting Started

Follow these steps to begin your assignment:

1. **Set up your project folder.** Create a well-organized directory structure for your code, data, and outputs.
2. **Explore the ACLED website.** Visit https://acleddata.com to familiarize yourself with the data and available resources.
3. **Read example analysis reports.** Browse ACLED Global Analysis for inspiration. As you read, consider the following guiding questions:
   - Which specific focus area interests you (e.g., geographic region, types of conflict, time period)?
   - What research questions about global conflict can be answered using the ACLED data?
   - What techniques covered in POLI3148 can help you address these research questions?
4. **Retrieve your data.** Download data of your choice using one of the following tools:
   - Data Export Tool
   - Download Data Files
5. **Understand the dataset.** Visit the ACLED Knowledge Base to learn how the dataset was constructed. In your notebooks, document the data generation process in two parts:
   - **(a)** The data generating process by ACLED, as described in their official documentation.
   - **(b)** Your own data retrieval process, including any filters or selections you applied.

Important Requirements

- **Dataset:** The primary dataset for this assignment **must** be ACLED. You may merge supplementary datasets if needed, but ACLED must be the main data source. Using a different dataset as the primary source will result in an automatic **Failure**, regardless of the quality of the output.
- **Programming Language:** The primary programming language for this assignment **must** be Python. Using any other programming language as your primary tool will result in an automatic **Failure**, regardless of the quality of the output.
- **AI Coding Assistant:** You are recommended to use **GitHub Copilot** within VS Code, as covered in the course. However, you may use another AI coding assistant of your choice. Whichever AI tools you use, you **must** clearly document them in your Note on AI Use.

Grading Rubric

Your assignment will be assessed on the following criteria:

Criterion	Description
Research Question	Whether you have posed important, insightful, and well-defined research question(s).
Methodological Rigor	Whether the project correctly and creatively applies data science tools, especially those covered in POLI3148.
Analytical Insight	Whether the data science methods generate important and interesting findings.
Presentation Quality	How well your Jupyter notebooks are documented; how effectively your interactive data-driven analysis report is presented as a coherent, polished product; the aesthetic quality of your data visualizations (proper labels, readable colors, clear legends); and whether figures and analysis text are effectively integrated.
Project Organization	Whether code, data files, and reports are properly organized and clearly documented within your project folder.

Academic Integrity

The instructor places the highest value on academic honesty. For this assignment, this specifically concerns the **independence and originality** of your work. All submitted work must be entirely your own independent, original work.

- **AI Use:** By design, the use of AI is integral to every component of this assignment. You are permitted and encouraged to use AI to assist with idea brainstorming, data retrieval, data cleaning, data visualization, data analysis, report generation, copy-editing, and other tasks. However, you are expected to use AI **wisely and responsibly**. First, AI cannot replace your own critical thinking. You must exercise your own analytical judgment to formulate valuable research questions and draw meaningful insights from the data. Second, you are responsible for **validating all AI-generated output**. This includes making your best effort to review AI-generated code to ensure it performs the analysis you intend, and carefully proofreading all text, including citations, whose final presentation has been processed through an AI assistant. AI hallucination can appear anywhere, and the responsibility for the accuracy and integrity of your final submission rests entirely with you.
- **Plagiarism:** If your output is highly similar to any publicly available content (e.g., webpages, research articles, blog posts), it will be considered plagiarism.
- **Collusion:** If your output is highly similar to that of one or more classmates, all students with similar output will be investigated and may be subject to disciplinary action.
- **Contract Cheating:** Having any third party complete the assignment on your behalf is strictly prohibited. If detected, the assignment will receive an automatic **F**, and the case will be referred for disciplinary action.

- **Verification:** The instructor may arrange a **Q&A session** to evaluate whether the assignment is your independent and original work. During this session, you will be asked detailed questions about how you completed the assignment, including your design process, intermediate outputs, and chat history with your AI assistant. You should be prepared to demonstrate and explain your work in full.

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